SISTEMAS DE APOIO À DECISÃO

SAD, EI 2026/2027



The project concept

APRESENTAÇÃO DO PROJETO

Concept

- Practical application of the concepts of software development and Business Intelligence
- Develop students' skills in leadership, management, mathematical modeling and programming
- Develop teamwork skills
- Resolution of concrete problems that typically arise in projects of this type in a real organization
- Stimulating the entrepreneurial capacity of students

- Motivation: https://www.dinheirovivo.pt/empresas/estudantes-da-universidade-do-porto-criam-solucoes-a-medida-para-empresas/

The project challenge

D4SP Decision Support System for Science Park Selection

APRESENTAÇÃO DO PROJETO

Motivation

- Choose the best science park
- Improve the probability of business success
- Improve profitability of a startup
- Difficulties in choosing:
  - What is the best science park for my startup?
  - What support will I have in the incubation process?
  - Which are the associated costs?
- Some useful links:
  - https://web.tecparques.pt
  - https://www.ani.pt/media/4382/caracterização-infraestruturas-tecnológicas_v4_2018.pdf
  - https://ec.europa.eu/regional_policy/sources/docgener/studies/pdf/stp_report_en.pdf

APRESENTAÇÃO DO PROJETO

Functional Requirements

- RF1. Authentication in the application
- RF2. Draw dashboard
- RF3. Introduce a new science park
- RF4 . Import science parks
- RF5. Search for science parks
- RF6. Perform a new simulation
  - Adoption of AHP method
  - Calculate operational costs
  - Show strong and weak points
- RF7. Export simulation results

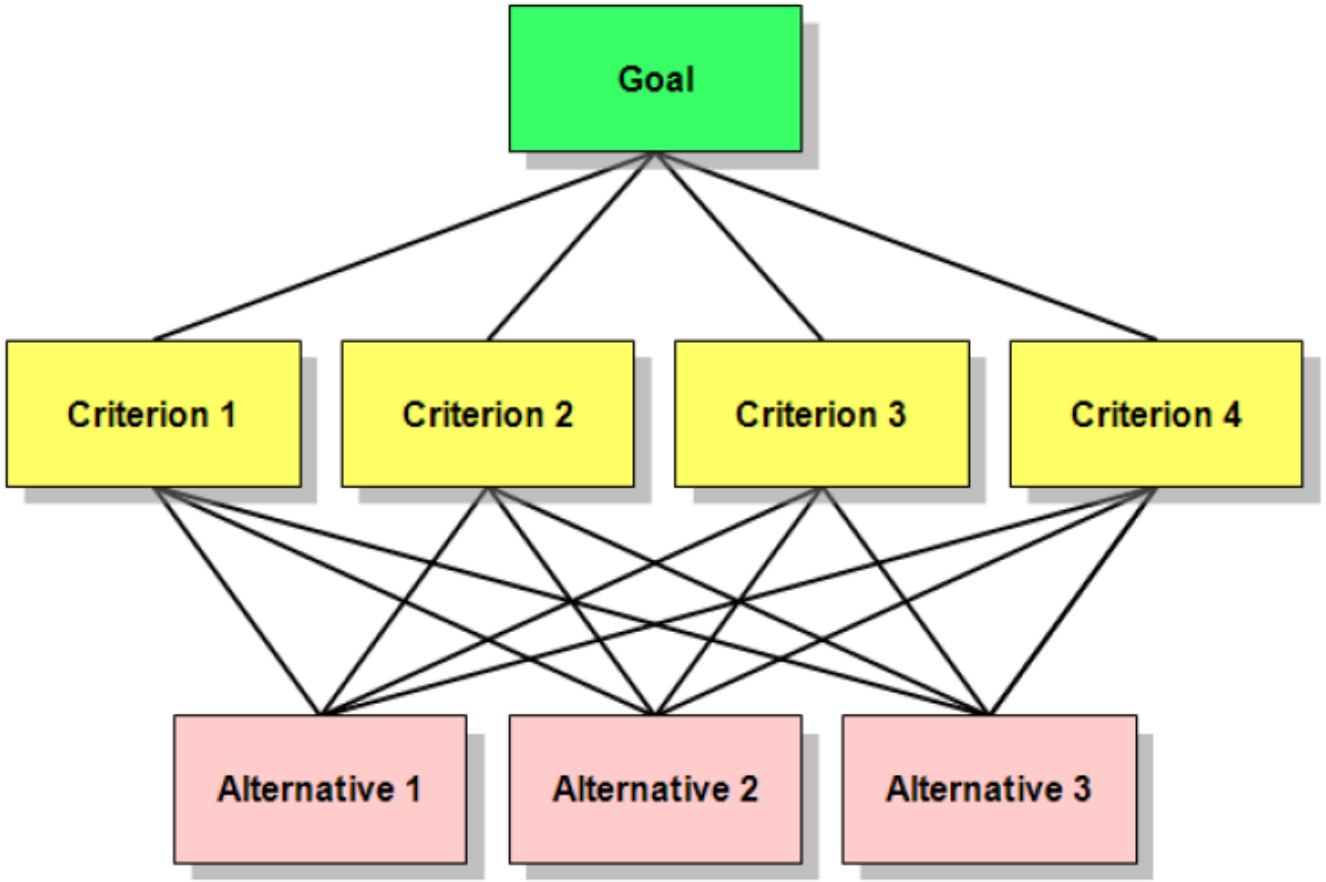
APRESENTAÇÃO DO PROJETO

Technologies to be adopted

- Visual Studio 2017/2019 C# or more recent
- Windows Presentation Foundation (WPF) / Windows Forms
- Language-Integrated Query (LINQ)
- SQL Server 2017 Express and SQL Server 2017 Management Studio or more recent
- XML, XSD, XML Schema Processor and DOM/SAX Parser

APRESENTAÇÃO DO PROJETO

Algorithms to be used - AHP method



APRESENTAÇÃO DO PROJETO

Algorithms to be used - AHP method

More information about AHP

- https://www3.diism.unisi.it/~mocenni/Note_AHP.pdf
- https://training.fws.gov/courses/references/tutorials/geospatial/CSP7306/Readings/AHP Technique.pdf
- http://www.rafikulislam.com/uploads/resourses/197245512559a37aadea6d.pdf
- https://core.ac.uk/download/pdf/80714029.pdf

Team Roles

APRESENTAÇÃO DO PROJETO

Project Manager

- Communication with the client and stakeholders
- Conflict management
- Project planning (GANTT + WBS)
- Performance report
- Evaluate the work team
- Final report and presentation
- Solidarity with all members of his/her team

APRESENTAÇÃO DO PROJETO

Analyst / System architect

- Creation of the SVN/GIT repository
- Requirements specification report
- Acceptance test plan
- Application installation process “easy-to-use”
- Installation and maintenance manual
- Participation in the code review process

APRESENTAÇÃO DO PROJETO

Programmer

- Software development
- Modeling and implementation of Business Intelligence (BI) algorithms
- Perform unit tests in the application
- Software configuration
- Implementation of the database in SQL Sever
- Implementation of graphical interfaces
- Participation and lead the code review process

APRESENTAÇÃO DO PROJETO

Secretary

- Implementation and maintenance of the company's website
- Elaborate the minutes of the meetings of the project
- Verify compliance with documentation standards
- Produce a cd/dvd with the project's documentation
- Produce a scientific paper to be published in an international journal or conference

Delivery Dates

Important note:

Sliding of the milestone in one working day implies a penalty of -1 value

APRESENTAÇÃO DO PROJETO

1ª Milestone – 12/11/2023

- Project planning (GANTT + WBS)
- Opening the web site
- Performance report I

APRESENTAÇÃO DO PROJETO

2ª Milestone – 17/12/2026

- Requirements specification report
- Acceptance tests report
- Code review I
- Survival test of the project
- Performance report II

APRESENTAÇÃO DO PROJETO

Last Milestone – January 2027 (07/01/2027)

- Evaluation and self-assessment of the project team
- Project's presentation
- Installation and maintenance manual
- Code review II
- Quantitative evaluation of the website
- Installation of the application "easy-to-use"
- CD/DVD/Pen Drive
- Scientific paper
- Final report of the project
- Performance report III

Evaluation

APRESENTAÇÃO DO PROJETO

Evaluation process

- **Evaluation:** 50% from professor + 50% from jury
- Evaluation carried out on the basis of the following elements:
  - Technical and scientific merit
  - Innovation level
  - Rigour in the process of modeling a real situation
  - Ability to work as a team and help each other
  - Quality of writing at spelling, syntax and semantics levels
  - Exposure capacity of the work

APRESENTAÇÃO DO PROJETO

Weights in the evaluation

Task	Value
Timeline (Gantt + WSB)	0,25
Performance Report	0,25
Final Report	3
Presentation	0,5
Website	1
Requirements Specification Report	1
Instalation Manual	0,25
Code Review	0,25
Acceptance Test Report	1
Graphical Interfaces + Technical Implementation	6
Business Intelligence Models	4
PEN	0,25
Scientific Paper	2

SISTEMAS DE APOIO À DECISÃO

Questões?



SISTEMAS DE APOIO À DECISÃO

